17. Using the following data, construct Fisher's Ideal index and show how it satisfies Factor Reversal Test and Time Reversal Test?

Commodity	Price in Rupees per unit		Number of units	
	Base year	Current year	Base year	Current year
A	6	10	50	56
B	2	2	100	120
C	4	6	60	60
D	10	12	50	24
E	8	12	40	36

18. Using Fisher's Ideal Formula, compute price index number for 1999 with 1996 as base year, given the following:

Year	Commodity: A		Commodity: B		Commodity: C	
	Price (Rs.)	Quantity (Kg)	Price (Rs.)	Quantity (Kg)	Price (Rs.)	Quantity (Kg)
1996	5	10	8	6	6	3
1999	4	12	7	7	5	4

19. Calculate Fisher's index number to the following data. Also show that it satisfies Time Reversal Test.

Commodity	2016		2017	
	Price (Rs.)	Quantity (Kg)	Price (Rs.)	Quantity (Kg)
Food	40	12	65	14
Fuel	72	14	78	20
Clothing	36	10	36	15
Wheat	20	6	42	4
Others	46	8	52	6

20. The following are the group index numbers and the group weights of an average working class family's budget. Construct the cost of living index number:

Groups	Food	Fuel and Lighting	Clothing	Rent	Miscellaneous
Index Number	2450	1240	3250	3750	4190
Weight	48	20	12	15	10

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